Nageeb Zaman

(Md Nageeb Bin Zaman)

Curriculum Vitae

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Research Interests

Scientific computing and high-performance/GPU computing; radiation hydrodynamics and radiative transfer; electronic-structure methods; stellar evolution; circumstellar interaction and mass loss in massive stars; superluminous and interacting supernovae.

Education

Ph.D., Physics (Theoretical Astrophysics), Louisiana State University, Baton Rouge, LA Expected May 2027
Research on hydrogen-poor interacting and superluminous supernovae, advised by Dr. Emmanouil Chatzopoulos. GPA 3.97/4.00. Passed the general exam (Ph.D. candidacy) in May 2026.

M.S., Physics, Central Michigan University, Mount Pleasant, MI 2021
Thesis: *Accelerating the Fermi-Löwdin Orbital Descriptor Optimization for Self-Interaction-Free DFT Calculations*, advised by Dr. Juan Peralta.

M.S., Physics, University of Dhaka, Dhaka, Bangladesh 2016
B.S., Physics, University of Dhaka, Dhaka, Bangladesh 2014

Research Experience

Graduate Researcher Jan 2022–Present
Louisiana State University, Supernovae and Mergers Group, Baton Rouge, LA
Advisor: Dr. Emmanouil Chatzopoulos. Research alongside teaching from January 2022; Graduate Research Assistant since August 2023.

- Built a multi-code simulation pipeline coupling binary stellar-evolution (MESA), 1D radiation-hydrodynamics (STELLA), and Monte Carlo radiative-transfer (SuperLite) codes to predict the time-dependent light output and spectra of stellar explosions.
- Designed and ran a grid of 30 circumstellar-interaction models (wind-like, compact-shell, and detached-shell structures, plus hydrogen-rich composition variants), mapping how density structure, radial extent, and composition shift peak luminosity, timescale, temperature, and spectral features (first-author paper under review, *ApJ*).
- Developed a joint diagnostic comparing luminosity, duration, temperature, and spectral evolution, and traced remaining model–data mismatches to specific physics limitations (LTE treatment, 1D geometry).
- Performed the spectroscopic analysis for a published study of rotating pulsational pair-instability supernovae: ran SuperLite Monte Carlo radiative-transfer calculations on HPC systems and compared synthetic spectra with observed supernovae using Duperfit, identifying the physical origin of time-varying spectral features (*ApJ*, 2025).
- Currently combining light-curve fitting (TigerFit) with physical models to constrain what powers individual superluminous supernovae.
- Developed Python pipelines (NumPy, SciPy, Pandas, Astropy) to process hundreds of GB of simulation output, extract physical parameters, and produce publication-quality figures.
- Ran large simulation campaigns totaling 300,000+ core-hours on Slurm-managed HPC clusters, using MPI-parallel codes; modified, compiled, and debugged Fortran and C/C++ scientific software.

Graduate Intern, Co-Design Summer School Summer 2023
CCS-7 (Applied Computer Science), Los Alamos National Laboratory, Los Alamos, NM

- One of a four-member team that built AresPK (originally Ares), a performance-portable multiphysics hydrodynamics code for thermonuclear (Type Ia) supernova explosions, extending the Parthenon adaptive-mesh-refinement (AMR) framework with Kokkos to run on both CPUs and GPUs.
- Implemented the nuclear-statistical-equilibrium (NSE) burning solver that computes energy release from thermonuclear burning; the team also added a monopole self-gravity solver and the first GPU implementation of the Helmholtz equation of state (via LANL's Singularity-EOS), which matched the reference implementation to machine precision.
- Ran strong- and weak-scaling studies on LANL's Chicoma supercomputer (AMD EPYC CPUs, NVIDIA A100 GPUs): perfect strong scaling up to 8 GPUs, one A100 GPU on average 34% faster than 64 CPU cores, and a largest run of a

1024^3 mesh on 512 MPI ranks (32,768 CPU cores). Presented at SC23 (poster) and the LANL summer-school exit talk.

Graduate Research Assistant 2019–2021
Central Michigan University, Computational Condensed Matter Physics Group, Mount Pleasant, MI
Advisor: Dr. Juan Peralta

- Modified and extended the Fortran-based FLO-SIC density functional theory (DFT) code to speed up self-interaction-corrected electronic-structure calculations of molecules.
- Implemented and benchmarked a Frozen Potential approximation that bypasses repeated potential evaluations, cutting force-evaluation runtime by 72–86% on test molecules (HCN, C_2H_4, CH_3F) while reproducing the exact forces' behavior near the optimal configuration; analyzed results in Python and visualized structures with VESTA. (M.S. thesis; talk at APS March Meeting 2021.)

Graduate Research Project, Statistical Physics 2015–2016
University of Dhaka, Department of Physics, Dhaka, Bangladesh
Supervisor: Prof. M. K. Hassan

- Co-developed a new percolation model based on random sequential ballistic deposition, in which particles dropped onto a square lattice bounce to a neighboring site when they land on an occupied one; wrote the lattice Monte Carlo simulation in C/C++ (unpublished).
- First research experience, covering Monte Carlo methods, scaling and critical phenomena, and scientific programming.

Publications

Under review

- **Zaman, N.**, & Chatzopoulos, E. "Photometric and Spectroscopic Diagnostics of Hydrogen-poor Circumstellar Interaction in Stripped-envelope Transients." *The Astrophysical Journal* (2026), first author. Code.

Published

- Huynh, T. N., Chatzopoulos, E., & **Zaman, N.** "Rotational Dynamics in Pulsational Pair-Instability Supernovae: Implications for Mass-Loss and Transient Events." *The Astrophysical Journal*, 991, 12 (2025). arXiv:2507.21307. (Co-author; spectroscopic analysis.)

Presentations

Diagnostics of Hydrogen-poor Circumstellar Interaction in Stripped-envelope Transients

- Oral, AAS 243rd Winter Meeting, New Orleans, LA (2024)
- Oral, Rise Time 2024: Explosive Astrophysics in the Era of High-Cadence Astronomy, Purdue University (2024)
- Poster, CraigFest 2024, UT Austin (2024)
- Poster, Transients from Space 2025, STScI / Johns Hopkins University (2025)

Ares – Simulating Type Ia Supernovae on Heterogeneous HPC Architectures

- Poster, SC23: International Conference for High Performance Computing, Denver, CO (2023)
- Exit talk, Co-Design Summer School, Los Alamos National Laboratory (2023)

Accelerating Fermi-Löwdin Orbital Descriptor Optimization for Self-Interaction-Free DFT

- Oral, APS March Meeting, virtual (2021)

Honors & Awards

Roussel Family Graduate Student Award, Dept. of Physics & Astronomy, Louisiana State University 2024
Best Teaching Assistant Award, Dept. of Physics, Central Michigan University 2020

Teaching Experience

Graduate Teaching Assistant LSU 2021–2023; CMU 2020–2021
Taught undergraduate physics laboratories; mentored students in experimental techniques, instrumentation, data analysis, and technical reporting. Prepared labs and demonstrations, troubleshot equipment, and graded reports and presentations.

Mathematics & Physics Teacher 2018–2019
Wordbridge School, Dhaka, Bangladesh. Taught mathematics and physics to middle-school students.

Technical Skills

Programming	Python, C/C++, Fortran, Bash, Java
Modeling Methods	Radiation hydrodynamics, Monte Carlo radiative transfer, stellar evolution, adaptive mesh refinement, DFT / electronic structure, light-curve and spectral fitting
Simulation Codes	MESA, STELLA, SuperLite, AresPK, AthenaPK / Parthenon, FLO-SIC, NRLMOL
HPC	Linux clusters, Slurm, MPI-parallel codes, CPU/GPU performance and scaling studies (NVIDIA A100); familiar with Kokkos, OpenMP
Data & Visualization	NumPy, SciPy, Pandas, Matplotlib, Astropy, Duperfit, TigerFit, MATLAB, Mathematica, Gnuplot, VESTA
Tools	Git/GitHub, VS Code, Vim, LaTeX

Workshops & Schools

Stellar Modeling for Nuclear Astrophysics Summer School, Louisiana State University 2022

Service & Leadership

Referee, *The Astrophysical Journal* 2024

Webmaster, LSU Physics Graduate Student Organization 2022–2023

Maintained the organization's website and supported communication and outreach for graduate students.